

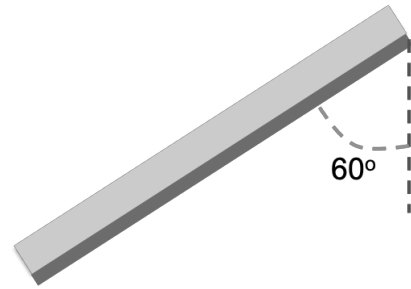
CHE 311 Homework Week 4

1. Falling film thickness

Please finish the calculation of the 3M tape film thickness using the following given parameters.

Consider the polymer solution a Newtonian fluid with viscosity $\mu = 2$ cp. The density of the fluid, ρ , is 1000 kg/m^3 . The dimension of the solid surface is 0.1 m wide and 1 m long. The tilted angle is 60° . The measured volumetric flow rate is 3 mL/s .

Please check Reynolds number at the end of your calculation to make sure the flow is Laminar.



2. Measure viscosity and viscosity

While wandering around the island of Hawaii during your winter break, you happen onto an active volcano eruption and observed a stream of hot lava flowing down a steeply inclined paved road (about 30° to the horizontal). You estimate the depth of the layer on the road as 10 cm , and using a tiny stone as a tracer, measure the surface velocity as 30 cm/s . Being eager to determine the density and viscosity of the lava, you toss a plank of wood (density 750 kg/m^3 and dimensions 5 cm (depth) $\times 1 \text{ m}$ (width) $\times 1 \text{ m}$ (length)) on top of the lava and note that it moves downstream at a velocity of 40 cm/s (before it goes up in flames).

Please decide the density and viscosity of the lava.

NOTE: the boundary conditions for the flow with the plank and the tiny stone is different. Since the stone is so small, its effect on the flow is negligible, and it is a no-shear stress boundary condition on top of the lava. The stone as a tracer only indicates the surface velocity of the lava. With the plank, the shear stress cannot be neglected, which is balanced with the gravity of the plank.

